Kali Linux Setup & OWASP Juice Shop Installation Guide

Pre-Requisites: Understanding Hypervisors & Installation

What is a Hypervisor?

A **hypervisor** is software that lets you run virtual computers (called virtual machines or VMs) inside your actual computer. Think of it like this:

- Your physical computer is like a building
- A hypervisor is like a property manager
- Virtual machines are like apartments inside the building

Each "apartment" (VM) acts like a separate computer with its own operating system, but they all share the building's resources (your computer's CPU, RAM, and storage). This lets you run Kali Linux safely alongside your regular operating system without affecting it.

Why use a hypervisor?

- Test software safely without risking your main computer
- Run multiple operating systems at the same time
- Easy to delete or reset if something goes wrong
- Great for learning and practicing cybersecurity

🪟 FOR WINDOWS USERS

Option 1: VirtualBox (FREE - Recommended for Beginners)

What is it? VirtualBox is a free, open-source hypervisor that works on almost any computer.

Download & Install:

1. **Download VirtualBox**
   - Visit: https://www.virtualbox.org/wiki/Downloads
   - Click "Windows hosts" under "VirtualBox Platform Packages"
   - Save the file to your Downloads folder
2. **Install VirtualBox**
   - Find the downloaded file (VirtualBox-x.x.x-Win.exe)

- Double-click to run the installer
- Click "Next" through all screens (default settings are fine)
- Click "Yes" if Windows asks for permission
- Click "Install"
- Wait for installation to complete
- Click "Finish"

3. **Verify Installation**
   - VirtualBox should open automatically
   - If not, press Windows key and type "VirtualBox"
   - Click to open it

Option 2: VMware Workstation Player (FREE for Personal Use)

What is it? VMware is professional virtualization software with a free version for personal, non-commercial use.

Download & Install:

1. **Download VMware Workstation Player**
   - Visit: https://www.vmware.com/products/workstation-player.html
   - Click "Download For Free"
   - Click "Download Now" for Windows
   - Save the file to your Downloads folder
2. **Install VMware Workstation Player**
   - Find the downloaded file (VMware-player-x.x.x.exe)
   - Double-click to run the installer
   - Click "Next"
   - Accept the license agreement
   - Click "Next" through all screens (default settings are fine)
   - **Uncheck** "Help improve VMware Workstation Player" if you prefer
   - Click "Install"

- Wait for installation to complete
- Click "Finish"
- Restart your computer if prompted

3. **First Launch**
   - Open VMware Workstation Player
   - Select "Use VMware Workstation Player for free for non-commercial use"
   - Click "Continue"
   - Click "Finish"

Option 3: Hyper-V Manager (FREE - Windows Pro/Enterprise/Education Only)

What is it? Hyper-V is Microsoft's built-in hypervisor that comes with Windows Pro, Enterprise, and Education editions.

Note: Hyper-V is NOT available on Windows Home edition.

Check if you have Windows Pro/Enterprise:

1. Press Windows key + R
2. Type `winver` and press Enter
3. Look at the Windows version shown
4. If it says "Home," you cannot use Hyper-V (use VirtualBox instead)

Enable Hyper-V (No Download Required):

Method 1: Using Windows Features

1. Press Windows key and type "Turn Windows features on or off"
2. Click the result
3. Scroll down and find "Hyper-V"
4. Check the box next to "Hyper-V" (this will check all sub-items)
5. Click "OK"
6. Wait for Windows to enable the feature
7. Click "Restart now" when prompted

Method 2: Using PowerShell (Faster)

1. Press Windows key and type "PowerShell"
2. Right-click "Windows PowerShell"
3. Select "Run as administrator"
4. Click "Yes" when asked for permission
5. Type or paste this command:

```powershell
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
```

6. Press Enter
7. Type Y and press Enter to restart

After Restart:

1. Press Windows key and type "Hyper-V Manager"
2. Click to open it
3. You're ready to use Hyper-V!

🍎 FOR macOS USERS

Option 1: VirtualBox (FREE - Recommended for Beginners)

What is it? VirtualBox is a free, open-source hypervisor that works on Mac.

Download & Install:

1. **Download VirtualBox**
   - Visit: https://www.virtualbox.org/wiki/Downloads
   - Click "OS X hosts" under "VirtualBox Platform Packages"
   - Save the file to your Downloads folder
2. **Install VirtualBox**
   - Find the downloaded file (VirtualBox-x.x.x-OSX.dmg)
   - Double-click to open it

- Double-click the VirtualBox.pkg icon
- Click "Continue" through the installer
- Click "Install"
- Enter your Mac password when prompted
- Wait for installation to complete
- Click "Close"

3. **Grant Permissions (Important!)**
   - Go to "System Preferences" (or "System Settings" on newer MacOS)
   - Click "Security & Privacy"
   - Click "General" tab
   - Click the lock icon and enter your password
   - You'll see a message about VirtualBox - click "Allow"
   - Restart your Mac
4. **Verify Installation**
   - Open Spotlight (Command + Space)
   - Type "VirtualBox"
   - Press Enter to open it

Option 2: VMware Fusion Player (FREE for Personal Use)

What is it? VMware Fusion is professional virtualization software with a free version for personal use.

Download & Install:

1. **Create a VMware Account (Required)**
   - Visit: https://www.vmware.com/products/fusion.html
   - Click "Register for a Personal Use License"
   - Create a free account
   - Check your email for the license key
2. **Download VMware Fusion Player**
   - Log into your VMware account

- Go to: https://www.vmware.com/products/fusion/fusion-evaluation.html
- Click "Download Now" for Mac
- Save the file to your Downloads folder

3. **Install VMware Fusion**
   - Find the downloaded file (VMware-Fusion-x.x.x.dmg)
   - Double-click to open it
   - Drag VMware Fusion to the Applications folder
   - Open Applications folder
   - Double-click VMware Fusion
   - Click "Open" if MacOS asks for confirmation
   - Enter your license key from email
   - Click "Continue"

4. **Grant Permissions**
   - MacOS will ask for various permissions
   - Click "OK" or "Allow" for each request
   - Go to System Preferences → Security & Privacy
   - Allow VMware Fusion in all requested areas
   - Restart your Mac if needed

Quick Comparison: Which Should You Choose?

Feature	VirtualBox	VMware	Hyper-V
Cost	Free	Free (Personal)	Free (Built-in)
Windows Home	✅ Yes	✅ Yes	❌ No
Windows Pro+	✅ Yes	✅ Yes	✅ Yes
MacOS	✅ Yes	✅ Yes	❌ No
Ease of Use	⭐⭐⭐⭐⭐	⭐⭐⭐⭐	⭐⭐⭐
Performance	⭐⭐⭐	⭐⭐⭐⭐	⭐⭐⭐⭐⭐
Best For	Beginners	Intermediate	Advanced/Windows Pro

Recommendation:

- **Beginners or Windows Home users:** VirtualBox
- **Mac users:** VirtualBox (easier) or VMware Fusion (better performance)
- **Windows Pro users:** Hyper-V (best performance, no download needed)

Part 1: Kali Linux Setup Options

Option 1: Setup Using Hyper-V

Prerequisites:

- Windows 10 Pro, Enterprise, or Education (or Windows 11)
- Virtualization enabled in BIOS
- Minimum 4GB RAM (8GB+ recommended)
- 20GB+ free disk space

Steps:

1. **Enable Hyper-V**
   - Open PowerShell as Administrator
   - Run: `Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All`
   - Restart your computer when prompted
2. **Download Kali Linux**
   - Visit: https://www.kali.org/get-kali/
   - Download the Hyper-V image (Pre-built VM)
   - Extract the downloaded .7z file
3. **Import Kali VM into Hyper-V**
   - Open Hyper-V Manager
   - Click "Import Virtual Machine" from the Actions menu
   - Browse to the extracted Kali folder
   - Select the folder containing the VM files
   - Choose "Copy the virtual machine" option
   - Complete the import wizard

4. **Configure VM Settings**
   - Right-click the imported VM → Settings
   - Adjust RAM (minimum 2GB, recommended 4GB+)
   - Configure network adapter (Default Switch or External)
   - Apply changes
5. **Start and Login**
   - Start the VM
   - Default credentials: `kali` / `kali`
   - Change password immediately after first login

Option 2: Setup Using Vagrant

Prerequisites:

- Vagrant installed (https://www.vagrantup.com/downloads)
- VirtualBox or VMware installed
- Command-line access

Steps:

1. **Create Project Directory**

```bash
mkdir kali-vagrant
cd kali-vagrant
```

2. **Initialize Vagrant with Kali**

```bash
vagrant init kalilinux/rolling
```

3. **Customize Vagrantfile (Optional)**
   - Open `Vagrantfile` in a text editor
   - Adjust settings as needed:

```ruby
Vagrant.configure("2") do |config|
  config.vm.box = "kalilinux/rolling"
  config.vm.provider "virtualbox" do |vb|
    vb.memory = "4096"
    vb.cpus = 2
    vb.gui = true
  end
end
```

4. **Start Vagrant VM**

```bash
vagrant up
```

5. **Access the VM**

```bash
# SSH access
vagrant ssh

# Or use the GUI if enabled
```

6. **Useful Vagrant Commands**

```bash
vagrant halt        # Stop the VM
vagrant suspend     # Suspend the VM
vagrant resume      # Resume suspended VM
vagrant destroy     # Delete the VM
vagrant reload      # Restart VM with new config
```

Option 3: Setup Using Docker

Prerequisites:

- Docker installed (https://docs.docker.com/get-docker/)
- Basic Docker knowledge

Note: Docker provides a lightweight Kali environment but with limitations compared to full VMs.

Steps:

1. **Pull Official Kali Docker Image**

```bash
docker pull kalilinux/kali-rolling
```

2. **Run Kali Container (Basic)**

```bash
docker run -ti kalilinux/kali-rolling /bin/bash
```

3. **Run with Persistent Storage**

```bash
# Create a volume for persistent data
docker volume create kali-data

# Run with mounted volume
docker run -ti -v kali-data:/root kalilinux/kali-rolling /bin/bash
```

4. **Run with Network Tools (Recommended)**

```bash
docker run -ti --privileged --network host \
  -v kali-data:/root \
  --name kali-container \
  kalilinux/kali-rolling /bin/bash
```

5. **Create a Custom Kali Image (Optional)**
   - Create a `Dockerfile`:

```dockerfile
```

```
FROM kalilinux/kali-rolling
RUN apt update && apt install -y \
    kali-linux-default \
    kali-tools-top10
```

- Build the image:

```bash
docker build -t my-kali .
docker run -ti my-kali /bin/bash
```

6. **Manage Docker Container**

```bash
# Start stopped container
docker start -ai kali-container

# Stop container
docker stop kali-container

# Remove container
docker rm kali-container
```

Part 2: Starting Kali Linux & Opening the Terminal

For First-Time Users: What is a Terminal?

A terminal (also called command line or console) is a text-based interface where you type commands to interact with your computer. Think of it as an alternative to clicking icons and menus - instead, you type instructions.

Option 1: Starting Kali in Hyper-V

Step 1: Open Hyper-V Manager

- Press the Windows key and type "Hyper-V"
- Click on "Hyper-V Manager"

Step 2: Start Your Kali VM

- In Hyper-V Manager, you'll see your Kali virtual machine in the list
- Right-click on the Kali VM
- Click "Connect" (this opens a window showing the VM)
- Click "Start" (green play button at the top)

Step 3: Log In to Kali

- Wait for Kali to boot up (you'll see a login screen)
- Username: `kali`
- Password: `kali` (you won't see characters as you type - this is normal for security)
- Press Enter

Step 4: Open the Terminal Once logged in, you'll see the Kali desktop. Here are three ways to open the terminal:

Method 1 (Easiest):

- Look at the top-left of the screen
- Click the terminal icon (looks like a black window with `>_` symbol)

Method 2:

- Click the "Applications" menu in the top-left corner
- Hover over "System Tools"
- Click "Terminal"

Method 3:

- Right-click anywhere on the desktop
- Select "Open Terminal Here"

Option 2: Starting Kali in Vagrant

Step 1: Open Your Command Prompt or Terminal

Windows:

- Press Windows key + R

- Type `cmd` and press Enter
- OR press Windows key, type "Command Prompt", and click it

Mac/Linux:

- Press Command + Space (Mac) or Ctrl + Alt + T (Linux)
- Type "terminal" and press Enter

Step 2: Navigate to Your Kali Directory

```bash
cd kali-vagrant
```

(This moves you into the folder where you set up Vagrant)

Step 3: Start Vagrant

```bash
vagrant up
```

(Wait for it to start - this may take a few minutes)

Step 4: Access Kali

If you enabled GUI in Vagrantfile:

- VirtualBox window will open automatically showing Kali desktop
- Log in with username: `vagrant` password: `vagrant`
- Open terminal using the methods described in the Hyper-V section above

If using SSH only (no GUI):

```bash
vagrant ssh
```

- You're now in the Kali terminal automatically!
- No GUI, just text commands

Option 3: Starting Kali in Docker

Step 1: Open Your Terminal/Command Prompt

- Follow the same steps as Vagrant Option above for your operating system

Step 2: Start Your Kali Container

If you created a named container:

```bash
docker start -ai kali-container
```

If starting fresh:

```bash
docker run -ti --privileged --network host kalilinux/kali-rolling /bin/bash
```

You're now in the Kali terminal!

- Docker Kali runs in terminal-only mode (no graphical interface)
- You'll see a prompt that looks like: `root@xxxxxx:/#`

Understanding the Terminal

What you'll see:

When you open a terminal, you'll see something like:

```
┌──(kali㉿kali)-[~]
└─$
```

OR

```
kali@kali:~$
```

This is called the "command prompt" - it's waiting for you to type a command.

Parts of the prompt:

- `kali@kali` - your username and computer name
- `~` - your current location (~ means home directory)
- `# Kali Linux Setup & OWASP Juice Shop Installation Guide

Part 1: Kali Linux Setup Options

Option 1: Setup Using Hyper-V

Prerequisites:

- Windows 10 Pro, Enterprise, or Education (or Windows 11)
- Virtualization enabled in BIOS
- Minimum 4GB RAM (8GB+ recommended)
- 20GB+ free disk space

Steps:

1. **Enable Hyper-V**
   - Open PowerShell as Administrator
   - Run: `Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All`
   - Restart your computer when prompted
2. **Download Kali Linux**
   - Visit: https://www.kali.org/get-kali/
   - Download the Hyper-V image (Pre-built VM)
   - Extract the downloaded .7z file
3. **Import Kali VM into Hyper-V**
   - Open Hyper-V Manager
   - Click "Import Virtual Machine" from the Actions menu
   - Browse to the extracted Kali folder
   - Select the folder containing the VM files
   - Choose "Copy the virtual machine" option
   - Complete the import wizard
4. **Configure VM Settings**

- Right-click the imported VM → Settings
- Adjust RAM (minimum 2GB, recommended 4GB+)
- Configure network adapter (Default Switch or External)
- Apply changes

5. **Start and Login**
   - Start the VM
   - Default credentials: `kali` / `kali`
   - Change password immediately after first login

Option 2: Setup Using Vagrant

Prerequisites:

- Vagrant installed (https://www.vagrantup.com/downloads)
- VirtualBox or VMware installed
- Command-line access

Steps:

1. **Create Project Directory**

```bash
mkdir kali-vagrant
cd kali-vagrant
```

2. **Initialize Vagrant with Kali**

```bash
vagrant init kalilinux/rolling
```

3. **Customize Vagrantfile (Optional)**
   - Open `Vagrantfile` in a text editor
   - Adjust settings as needed:

```ruby
```

```
Vagrant.configure("2") do |config|
  config.vm.box = "kalilinux/rolling"
  config.vm.provider "virtualbox" do |vb|
    vb.memory = "4096"
    vb.cpus = 2
    vb.gui = true
  end
end
```

4. **Start Vagrant VM**

```bash
vagrant up
```

5. **Access the VM**

```bash
# SSH access
vagrant ssh

# Or use the GUI if enabled
```

6. **Useful Vagrant Commands**

```bash
vagrant halt        # Stop the VM
vagrant suspend     # Suspend the VM
vagrant resume      # Resume suspended VM
vagrant destroy     # Delete the VM
vagrant reload      # Restart VM with new config
```

Option 3: Setup Using Docker

Prerequisites:

- Docker installed (https://docs.docker.com/get-docker/)
- Basic Docker knowledge

Note: Docker provides a lightweight Kali environment but with limitations compared to full VMs.

Steps:

1. **Pull Official Kali Docker Image**

```bash
docker pull kalilinux/kali-rolling
```

2. **Run Kali Container (Basic)**

```bash
docker run -ti kalilinux/kali-rolling /bin/bash
```

3. **Run with Persistent Storage**

```bash
# Create a volume for persistent data
docker volume create kali-data

# Run with mounted volume
docker run -ti -v kali-data:/root kalilinux/kali-rolling /bin/bash
```

4. **Run with Network Tools (Recommended)**

```bash
docker run -ti --privileged --network host \
  -v kali-data:/root \
  --name kali-container \
  kalilinux/kali-rolling /bin/bash
```

5. **Create a Custom Kali Image (Optional)**
   - Create a `Dockerfile`:

```dockerfile
```

```
FROM kalilinux/kali-rolling
RUN apt update && apt install -y \
    kali-linux-default \
    kali-tools-top10
```

- Build the image:

```bash
docker build -t my-kali .
docker run -ti my-kali /bin/bash
```

6. **Manage Docker Container**

```bash
# Start stopped container
docker start -ai kali-container

# Stop container
docker stop kali-container

# Remove container
docker rm kali-container
```

- indicates you're a regular user (`#` means you're root/admin)

Basic Terminal Tips:

1. **Type carefully** - Spelling matters! `Update` is different from `update`
2. **Copy and Paste:**
   - **Copy:** Ctrl + Shift + C (in terminal) or select text and right-click → Copy
   - **Paste:** Ctrl + Shift + V (in terminal) or right-click → Paste
3. **Cancel a command:**
   - Press `Ctrl + C` to stop a running command
4. **Clear the screen:**
   - Type `clear` and press Enter
5. **Command history:**
   - Press ↑ (up arrow) to see previous commands

- Press ↓ (down arrow) to go forward through history

6. **Auto-complete:**
   - Type the first few letters and press Tab to auto-complete file/folder names

---

Basic Navigation Commands (Try These!)

```bash
# See where you are
pwd

# List files in current folder
ls

# List files with details
ls -la

# Go to home directory
cd ~

# Go to a specific folder
cd Downloads

# Go back one folder
cd ..

# Create a new folder
mkdir my-folder

# See what's running
top
# (Press 'q' to quit)
```

Testing Your Terminal

Once your terminal is open, try typing this command and press Enter:

```bash
whoami
```

You should see your username (probably `kali`). If you see this, your terminal is working!

Try another:

```bash
ls
```

This lists the files in your current location.

Congratulations! You're now ready to use the terminal for the OWASP Juice Shop installation below.

Part 3: Post-Installation Setup & OWASP Juice Shop

Prerequisites

- Kali Linux installed and running (using any method above)
- Terminal open (see Part 2 above)
- Internet connection
- Root or sudo access

Step 1: Update Kali Linux

Update package lists and upgrade system:

```bash
sudo apt update
sudo apt upgrade -y
sudo apt dist-upgrade -y
```

Clean up unnecessary packages:

```bash
sudo apt autoremove -y
sudo apt autoclean
```

Reboot (if kernel was updated):

```bash
sudo reboot
```

Step 2: Install Required Dependencies

Install Node.js and npm (required for OWASP Juice Shop):

```bash
# Install Node.js from NodeSource repository
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs

# Verify installation
node --version
npm --version
```

Install Git (if not already installed):

```bash
sudo apt install -y git
```

Step 3: Install OWASP Juice Shop

Method A: From Source (Recommended)

1. **Clone the repository:**

```bash
cd ~
git clone https://github.com/juice-shop/juice-shop.git
cd juice-shop
```

2. **Install dependencies:**

```bash
npm install
```

3. **Start Juice Shop:**

```bash
npm start
```

4. **Access the application:**
   - Open browser and navigate to: `http://localhost:3000`

Method B: Using Docker (if Docker is available)

```bash
docker pull bkimminich/juice-shop
docker run -d -p 3000:3000 --name juice-shop bkimminich/juice-shop
```

Access at: `http://localhost:3000`

Method C: Using npm package (Quick Install)

```bash
npm install -g juice-shop
juice-shop
```

Step 4: Running and Managing OWASP Juice Shop

Start Juice Shop (from source directory):

```bash
cd ~/juice-shop
npm start
```

Run in background:

```bash
nohup npm start &
```

Stop Juice Shop:

```bash
# If running in foreground: Press Ctrl+C

# If running in background:
pkill -f "npm start"
# or
ps aux | grep juice-shop
kill [PID]
```

Update Juice Shop to latest version:

```bash
cd ~/juice-shop
git pull
npm install
npm start
```

Step 5: Configure Juice Shop (Optional)

Create custom configuration:

```bash
cd ~/juice-shop
cp config/default.yml config/custom.yml
```

Edit custom configuration:

```bash
nano config/custom.yml
```

Run with custom config:

```bash
NODE_ENV=custom npm start
```

Step 6: Useful Commands

Check if Juice Shop is running:

```bash
netstat -tulpn | grep 3000
# or
lsof -i :3000
```

View logs:

```bash
cd ~/juice-shop
tail -f nohup.out
```

Reset Juice Shop database:

```bash
cd ~/juice-shop
rm data/juiceshop.sqlite
npm start
```

Troubleshooting

Port already in use:

```bash
# Find process using port 3000
sudo lsof -i :3000
# Kill the process
sudo kill -9 [PID]
```

npm installation errors:

```bash
```

```
# Clear npm cache
npm cache clean --force
# Remove node_modules and reinstall
rm -rf node_modules
npm install
```

Permission errors:

```bash
# Fix npm permissions
sudo chown -R $USER:$USER ~/.npm
sudo chown -R $USER:$USER ~/juice-shop
```

Additional Tips

1. **Create a startup script:**

```bash
nano ~/start-juiceshop.sh
```

Add:

```bash
#!/bin/bash
cd ~/juice-shop
npm start
```

Make executable:

```bash
chmod +x ~/start-juiceshop.sh
```

2. **Access from other machines on your network:**
   - Start Juice Shop with host binding:

```bash
```

```
HOST=0.0.0.0 npm start
```

- Access via: `http://[kali-ip]:3000`

3. **Keep Kali and tools updated regularly:**

```bash
sudo apt update && sudo apt upgrade -y
```

Quick Reference Commands

```bash
# System Update
sudo apt update && sudo apt upgrade -y

# Start Juice Shop
cd ~/juice-shop && npm start

# Update Juice Shop
cd ~/juice-shop && git pull && npm install

# Check Juice Shop status
netstat -tulpn | grep 3000

# Stop all node processes
pkill -f node
```